

Collision early warning technology of transformer bushing lifting based on BIM

Jiefeng Zhang, Helong Li, Guangze Zhu*, Nan Hong, Sheng Hua, Zhouyang Tan, Boming Li, Li Chen
Zhejiang Power Transmission and Transformation Engineering Co., Ltd., Hangzhou, Zhejiang, China

*Corresponding author's e-mail: 1109947846@qq.com

Abstract: Transformer bushing hoisting is a high-risk and critical process in substation engineering, especially in densely structured indoor or expansion stations. Traditional planning methods that rely on two-dimensional drawings and experience are difficult to cope with complex spatial conflicts in dynamic processes. This article proposes a dynamic collision warning technology that integrates Building Information Modeling (BIM) and physical engine simulation, aiming to achieve precise pre-simulation and real-time warning during the lifting process, providing a systematic solution for digital control of large-scale equipment lifting under complex working conditions.

Keywords—transformer hoisting; BIM; physical engine; collision detection

I. INTRODUCTION

As power systems evolve toward higher voltages, greater capacities, and more compact designs, the construction environment for substation projects has become increasingly complex. Transformers, as core equipment in substations, require specialized installation procedures—particularly for high-voltage bushing hoisting. These operations, characterized by massive components, stringent precision requirements, and often limited working space, rank among the most technically challenging and high-risk processes in project implementation [1-4]. Traditional hoisting design approaches primarily rely on 2D blueprints, engineers' experience, and simplistic 3D visualizations, lacking precise quantitative analysis of four-dimensional interactions between hoisting systems and surrounding environments [5-7]. This static, discrete planning model exhibits three critical limitations: First, it fails to accurately assess dynamic interference between hoisting arm movements (pitching, rotation, wire rope oscillation) and static obstacles. Second, it cannot effectively predict potential collisions caused by minor object movements induced by wind forces, operational inertia, or mechanical flexibility. Third, the lack of intuitive risk communication during design briefings leaves field operators inadequately aware of hazards, frequently resulting in emergency stoppages, component abrasions, or major safety incidents that cause economic losses and project delays.

To address these challenges, Building Information Modeling (BIM) technology has been progressively adopted in power engineering construction due to its advantages in 3D visualization, information integration, and collaborative management [8-11]. Current research and practices primarily

focus on using BIM for static collision detection and construction process visualization simulations. However, such applications essentially discretize dynamic processes into keyframes for static interference analysis, failing to incorporate temporal continuity and physical laws. This approach cannot authentically simulate the dynamic behavior and continuous collision responses during hoisting operations [9-17]. In recent years, physics engine technologies originating from computer graphics and gaming industries have emerged as a new technical pathway for immersive, high-fidelity construction simulations. These engines leverage efficient rigid-body dynamics calculations, Continuous Collision Detection (CCD), and real-time simulation capabilities [18-24]. By simulating object motion under gravity, constraints, and external forces based on Newtonian mechanics, physics engines perform frame-by-frame contact and collision calculations—a methodology that aligns perfectly with the dynamic characteristics of hoisting operations.

This paper proposes an innovative technical integration approach by deeply integrating BIM's high-precision geometric and information models with the dynamic simulation and real-time collision detection capabilities of physical engines to develop a dynamic collision early-warning system for transformer bushing hoisting. The research aims to transcend traditional static interference inspections by creating a computable virtual hoisting environment. Within this environment, hoisting plans can undergo trial runs based on physical laws, enabling precise identification, quantification, and avoidance of all static and dynamic risks before construction. Ultimately, this establishes a standardized technical process encompassing model creation, simulation setup, dynamic analysis, and early-warning output, thereby providing scientific decision-making support and proactive safety barriers for critical hoisting operations in complex spaces.

II. SYSTEM'S OVERALL TECHNICAL ARCHITECTURE AND KEY MODULES

The collision warning technology system proposed in this study is a closed-loop process from data to decision-making, with its core being the construction of a virtual lifting simulation environment based on a digital twin. The technical architecture of the entire system follows the logical hierarchy of "data layer-model layer-simulation layer-application layer", as shown in Figure 1.

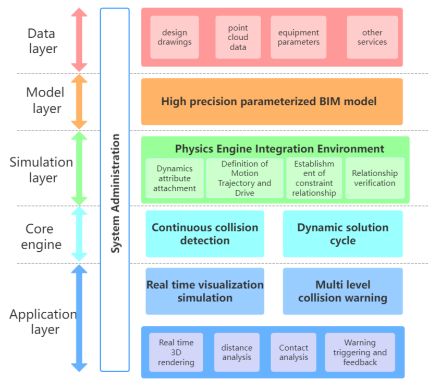


Figure 1. Architecture of Collision Warning System for Transformer Bushing Hoisting Based on BIM and Physical Engine.

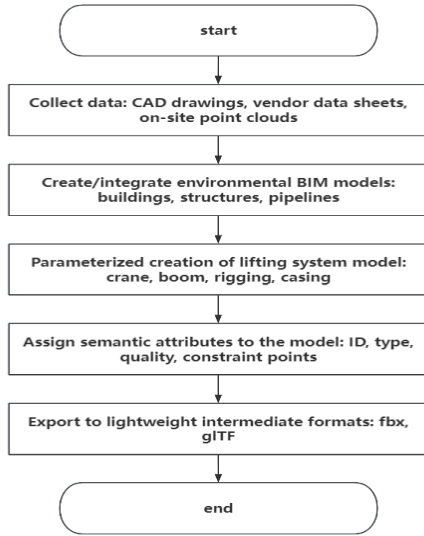


Figure 2. High-precision BIM Modeling and Data Processing Flowchart.

A. Construction of high-precision parametric BIM model

A comprehensive BIM model with a high level of detail is developed to cover all elements, including lifting environments, objects, machinery, and rigging. The key lies in parametric and semantic design: the model should not only include geometric shapes but also embed physical property parameters and motion relationship parameters. The model's precision must reach the level required for construction plan analysis. For example, steel cables should be modeled as flexible bodies to simplify chain models into non-linear configurations, as shown in Figure 2.

B. Integration of Physics Engine and Dynamics Simulation Settings

This module serves as the system's brain. It imports the processed BIM model into the physics engine. The key steps include the following:

1. Rigid body properties: A rigid body is created for each moving mI part, and its mass and inertia tensor are assigned. The inertia tensor is crucial for irregularly shaped components and is typically calculated by the engine using a geometry approximation or defined manually.

$$I = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \quad (1)$$

I represents the inertia tensor of a rigid body, which is a 3x3 matrix that describes the inertia characteristics of a rigid body rotating around different axes. The diagonal elements I_{xx}, I_{yy}, I_{zz} of the matrix represent the moment of inertia of the rigid body rotating around the x-axis, y-axis, and z-axis, respectively. Non-diagonal elements (such as I_{xy}) are called inertial products, representing the inertial coupling of rigid bodies in different directions.

2. Constraint and Joint Setup: The engine's Joint function simulates real mechanical connections. For example, the boom segments use hinge joints for pitch movement, the rotary platform and chassis use rotational joints, and the wire rope and hook use ball-and-socket joints. The stiffness and damping parameters of constraints must be calibrated according to the mechanical manual.

3. Motion Drive and Control Logic: The crane's operational logic is defined through script code. For example, the boom's elevation $\theta(t)$ angle is adjusted by controlling the motor torque or angular velocity of the hinge joint, while the crane's rotation is simulated by regulating the speed of the swivel joint. The motion trajectory can be either a predefined path or a real-time human-machine interaction input. Below is a simplified pseudocode logic.

C. Continuous Collision Detection and Warning Mechanism

At each simulation time step, the Δt physics engine performs dynamic calculations and collision detection. The system employs a pairwise detection strategy, focusing on interference between the "lifting system set" and the "environmental obstacle set". The core algorithm combines the bounding box hierarchy (BVH) with either the Separated Axis Theorem (SAT) or Gilbert-Johnson-Keerthi (GJK) algorithm to balance efficiency and accuracy.

The warning mechanism is based on the minimum Euclidean distance d calculated in real time. The system preset two thresholds: the warning distance D_{warn} and the danger distance D_{danger} . When $d \leq D_{warn}$, the system highlights the components about to interfere in the visualization interface and records logs; when $d \leq D_{danger}$, an emergency warning is triggered, and the simulation can be automatically paused. The calculation of distance d can be simplified to the distance between the nearest feature points of the two objects, provided by the query function of the physics engine.

Through this architecture, the system has achieved a leap from a static BIM model to a dynamic, interactive, and physically realistic simulation warning platform, providing a reliable technical environment for subsequent solution optimization and decision-making.

III. COLLISION DETECTION ALGORITHM OPTIMIZATION AND KEY PARAMETER CALIBRATION

To achieve high-precision collision warnings while ensuring simulation real-time performance, this system has optimized the physics engine's collision detection workflow

and performed fine-tuned calibration of key physical parameters. The core solution involves developing an efficient hierarchical collision detection framework, coupled with engineering-calibrated dynamic parameters for the simulation model, effectively bridging the gap between virtual simulations and physical reality.

A. Optimization of Layered Collision Detection Algorithm

The computation of the triangle patch detection of all models is very large. The optimization process for layered collision detection is shown in Figure 3. Therefore, the system adopts the classic three-stage hierarchical detection strategy: Broadphase, Narrowphase, and precise contact generation.

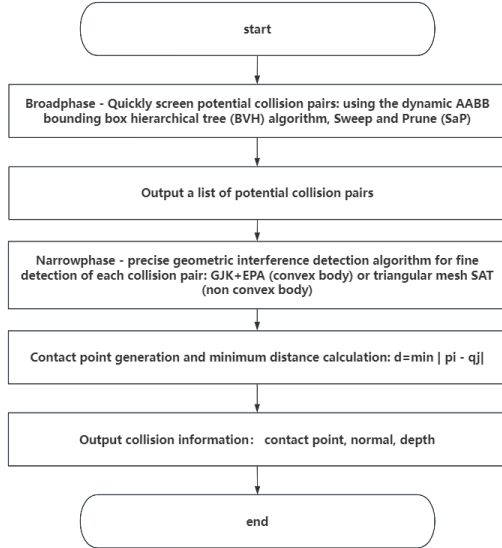


Figure 3. Flowchart of hierarchical collision detection optimization.

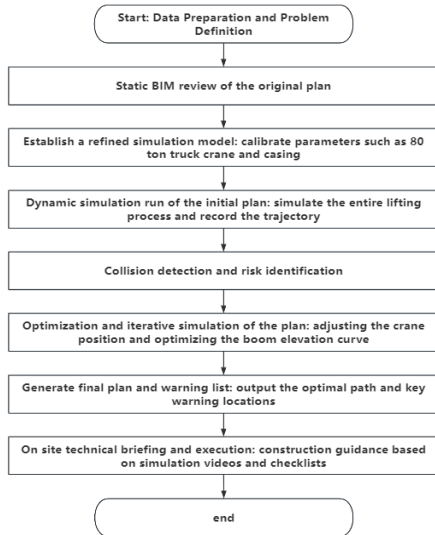


Figure 4. Flowchart of engineering implementation.

During the Broadphase stage, the system maintains an axis-aligned bounding box (AABB) for each moving object and employs a "sweep and prune" algorithm to quickly eliminate pairs of objects that are clearly non-intersecting in

space, reducing the detection complexity from $O(n^2)$ to approximately $O(n \log n)$. In the Narrowphase stage, the Gilbert-Johnson-Keerthi (GJK) algorithm is applied to determine convex hull intersections for filtered object pairs. If an intersection is detected, the Expanding Polytope Algorithm (EPA) calculates penetration depth and contact normals; for non-convex hulls, the objects are decomposed into combinations of convex hulls or processed using conservative forward mesh detection. The system specifically designs simplified collision bodies (such as capsule bodies and composite convex hulls) for steel wire ropes (discretized as multiple short cylindrical rigid bodies connected by ball joints) and porcelain bushings (complex curved surfaces), significantly enhancing computational efficiency while maintaining physical realism.

B. Calibration of Key Physical Parameters and Model Validation

The credibility of the simulation depends on the accuracy of the input parameters. The system calibrates the parameters by the following methods:

1. Mass and Inertia Properties: The equipment mass m_{lp} is directly obtained from manufacturer documentation. For irregular components, the inertia tensor is estimated by combining the BIM model geometry with density or simplified using the parallel axis theorem. For example, the moment of inertia around the central axis of a boom joint can be approximated as:

$$I_{xx} \approx \frac{m}{12}(h^2 + d^2) \quad (2)$$

The height h and width of the section are denoted as h and w , respectively, and m represents the mass of the boom joint.

2. Parameter Calibration for Constraints and Drives: The ck joint's friction coefficient, damping coefficient, and stiffness coefficient are determined by consulting mechanical manuals or performing basic "digital twin calibration". For instance, by driving a virtual crane to execute a predefined simple no-load motion, the joint parameters are adjusted to align the simulated motion curve with actual mechanical characteristic data or operational manual specifications.

3. Modeling of environmental force field: To simulate the influence of gusts, a force varying $F_{wind}(t)$ with time and space can be applied to the rigid casing, and its magnitude is calculated based on the simplified Bernoulli equation:

$$F_{wind} = \frac{1}{2} \cdot C_d \cdot \rho_{air} \cdot A \cdot v(t)^2 \quad (3)$$

The parameter C_d is the drag coefficient, A is the windward area, and $v(t)$ is the time-varying wind speed. The system supports the input of wind spectrum data or simplifies it into impulse loads for the most unfavorable condition test.

Through the algorithm optimization and parameter calibration, the system achieves a balance between the calculation efficiency and the simulation fidelity, and provides a reliable technical kernel for dynamic collision warning.

IV. ENGINEERING APPLICATION EXAMPLES AND EFFECT VERIFICATION

To verify the practicability and effectiveness of the proposed technology, this study applied it to the expansion project of the third main transformer at 500 kV Hemu Substation. The substation has a complex structure, with multi-layer steel frames above the hoisting area, dense busbars and grounding wires on the sides and rear, resulting in extremely narrow working space. Traditional methods pose extremely high risks.

A. Implementation Process of the Practical Application

The implementation follows the systematic process shown in Figure 4.

The procedure involves first constructing a high-precision BIM environment that incorporates all obstacles using design drawings and on-site laser-scanned point clouds. Then, based on the preliminary hoisting plan, the system establishes the crane's initial position and outrigger locations while mapping the casing's lifting, translation, flipping, and positioning trajectories. The initial dynamic simulation not only confirmed known static risk points but also identified two critical new dynamic risks:

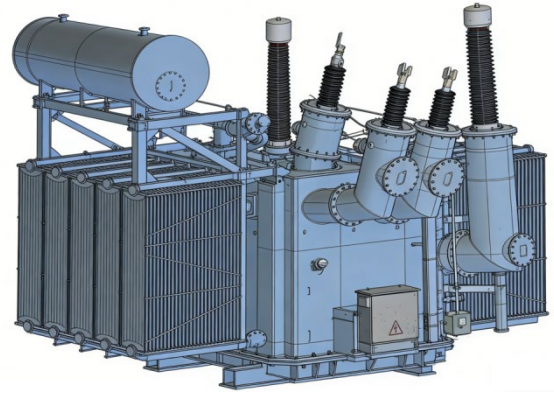
The sleeve, after crossing the lower steel beam, may experience slight pendulum-like motion due to wire rope torsion, potentially causing its skirt edge to scrape against the upper slant support angle steel at $t = 142\text{ s}$, with a minimum predicted distance $d_{\min} = 3.8\text{ cm}$;

When the boom rotates to a certain angle, its head pulley group is too close to the tubular busbar structure $d_{\min} = 5.1\text{ cm}$ on one side, which increases the risk under wind force.

The physical image of the transformer is shown in Figure 5, and its detailed modeling is shown in Figure 6.



Figures 5. Physical Image of Transformer.



Figures 6. Fine Modeling of Transformer

Based on the early warning system, the project team conducted multiple rounds of simulation optimization: they moved the crane 1.5 meters outside the station and adjusted the initial boom elevation angle from 68° to 72° . The optimized simulations demonstrated that all dynamic risk points were eliminated, with the minimum safety distance increased to over 15 cm. The system generated operational manuals and simulation videos containing key node coordinates, lifting action instructions, and warning zone coordinates. The generated 3D animation of the lifting operation based on BIM model rendering is shown in Figure 7.



Figure 7. 3D animation of hoisting operation based on BIM model rendering

B. Effect Comparison Analysis and Verification

The comparison of key indicators before and after the application of technology is shown in Table 1. By comparing the indices before and after the application of this technology, the effect is remarkable.

The actual hoisting process at the site closely matched the simulation. When the casing passed through the first original risk zone, the monitoring team confirmed that its swinging trajectory aligned with the simulation, while the optimized plan maintained sufficient safety clearance. Ultimately, the casing was installed safely, precisely, and efficiently without any scraping incidents.

Table 1. Comparison of Key Indicators Before and After Technology Application.

	Traditional solution design	Early Warning Technology Based on BIM and Physical Engine	Improvement effect
Risk identification	There are 3 static interference points.	3 static points, 2 dynamic risk points	Increase by 40%
Optimization of the scheme	Depends on experience and multiple site surveys, approximately 5-7 days	Complete rapid iteration in a virtual environment within 1-2 days	Increase by 70%
Expression form	2D drawings and text descriptions	3D Dynamic Simulation Video and Visualized Early Warning Report	Significantly improve
Lifting process	High alert. A dedicated person is required to monitor multiple locations and make minor adjustments.	Follow the preview steps. The process is smooth, and you can set it up in one go.	Time reduced by 35%, success rate 100%
Security Management	Passive response, relying on personnel vigilance	Active early warning, pre-emptive risk quantification, and mitigation	Significantly improve

V. CONCLUSION AND PROSPECT

To address the high-risk management challenges in transformer bushing hoisting operations within complex spatial environments, this study proposes and establishes a collision early-warning technical framework that integrates BIM high-precision modeling technology with physics engine-based dynamic simulation capabilities. By converting construction elements into digital models with realistic physical attributes, the system replicates the entire hoisting process dynamics in virtual environments, achieving a technological leap from static interference inspections to real-time collision warnings. Engineering case studies demonstrate that the system effectively identifies dynamic oscillation and spatiotemporal interference risks undetectable by traditional methods. Through pre-hoc optimization, it quantifies safety risks and preemptively mitigates them, significantly enhancing the scientific rigor of hoisting plans, first-time success rates, and construction safety while delivering substantial economic and schedule benefits. This research provides a practical solution with significant value for digital and refined management of high-risk operations in the power construction fields.

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